Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/14/2007
PI Name:	Schiffer, Peter Ph.D.		
Project Title:	Correlation Effects in Dense Wet and Dry Granular Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	schiffer@phys.psu.edu	**Fax:**	FY 814-865-3604
PI Organization Type:	UNIVERSITY	**Phone:**	814-865-5982
Organization Name:	Pennsylvania State University		
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PI Address 2:			
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City:	University Park	**State:**	PA
Zip Code:	16802	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
Contact Monitor:	Agui, J. H.	**Contact Phone:**	216-433-5409
Contact Email:	Juan.H.Agui@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA33G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This grant is supporting research into the fundamental properties of granular media. There is a particular focus on the properties of dense granular media and the development of correlations therein. A better understanding of the development and evolution of correlations in dense granular media. Granular media are used in a wide range of industrial processes, and understanding their properties is important to improving these processes. NASA missions (such as setting up lunar or martian bases) will also require extensive handling of granular materials, and therefore the understanding of these materials will be important for the design and implementation of those missions. Finally, the understanding of granular materials is of great fundamental scientific		

	importance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	More efficient industrial processes and better understanding of fundamental science always improves life on earth. This research should yield advances in both.
Task Progress:	This research program supported a range of experiments on the physics of granular materials. The phenomena studied included local jamming (probed by measuring thee granular penetration force from both above and below), the effects of thermal cycling on granular materials, and the nature of granular flux from the highly fluidized granular medium. All of these experiments have yielded interesting results which have advanced the fundamental understanding of granular materials and are relevant to both lunar and Martian missions. Peer-reviewed publications resulting from this research are listed in the Bibliography.
Bibliography Type:	Description: (Last Updated: 11/15/2007)
Articles in Peer-reviewed Journals	Chen K, Cole J, Conger C, Draskovic J, Lohr M, Klein K, Scheidemantel T, Schiffer P. "Granular materials: packing grains by thermal cycling." Nature. 2006 Jul 20;442(7100):257. PMID: 16855580 , Jul-2006
Articles in Peer-reviewed Journals	Chen K, Stone MB, Barry R, Lohr M, McConville W, Klein K, Sheu BL, Morss AJ, Scheidemantel T, Schiffer P. "Flux through a hole from a shaken granular medium." Phys Rev E Stat Nonlin Soft Matter Phys. 2006 Jul;74(1 Pt 1):011306. PMID: 16907088 , Jul-2006
Articles in Peer-reviewed Journals	Schiffer P. "Granular physics: a bridge to sandpile stability." Nature Physics 2005 Oct;1(1):21-2. http://dx.doi.org/10.1038/nphys129 , Oct-2005
Articles in Peer-reviewed Journals	MB, Barry R, Bernstein DP, Pelc MD, Tsui YK, Schiffer P. "Local jamming via penetration of a granular medium." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Oct;70(4 Pt 1):041301. PMID: 15600401 , Oct-2004
Articles in Peer-reviewed Journals	Stone MB, Bernstein DP, Barry R, Pelc MD, Tsui YK, Schiffer P. "Stress propagation: getting to the bottom of a granular medium" Nature. 2004 Feb 5;427(6974):503-4. PMID: 14765185 , Feb-2004